

Module 04 Lab - Exploratory Data Analysis **Objective:** To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model. **This lab is fully coded.** Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading ****What is Exploratory Data Analysis (EDA)?**** EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data. What stories does it tell? Are there errors or missing values? Are there strong relationships between variables? EDA helps answer these questions. We will use the famous Titanic dataset for this lab. It contains information about passengers and, crucially, whether they survived the disaster.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the Titanic dataset
df= pd.read_csv(
    "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"
)

print("--- First 5 Rows ---")
print(df.head())

print("--- Basic Info ---")
df.info()
```

```
--- First 5 Rows ---
   PassengerId  Survived  Pclass \
0             1         0       3
1             2         1       1
2             3         1       3
3             4         1       1
4             5         0       3

   Name                               Sex  Age  SibSp \
0  Braund, Mr. Owen Harris             male  22.0     1
1  Cumings, Mrs. John Bradley (Florence Briggs Th... female  38.0     1
2    Heikkinen, Miss. Laina             female  26.0     0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)   female  35.0     1
4    Allen, Mr. William Henry             male  35.0     0

   Parch  Ticket   Fare Cabin Embarked
0      0   A/5 21171   7.2500   NaN      S
1      0    PC 17599  71.2833   C85      C
2      0  STON/O2. 3101282   7.9250   NaN      S
3      0   113803   53.1000  C123      S
4      0  373450   8.0500   NaN      S

--- Basic Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   PassengerId     891 non-null   int64
1   Survived        891 non-null   int64
2   Pclass          891 non-null   int64
3   Name            891 non-null   object
4   Sex             891 non-null   object
5   Age            714 non-null   float64
6   SibSp           891 non-null   int64
```

```

7 Parch      891 non-null  int64
8 Ticket     891 non-null  object
9 Fare       891 non-null  float64
10 Cabin     204 non-null  object
11 Embarked   889 non-null  object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

```

Part 2: Descriptive Statistics Let's start by getting a high-level numerical summary of the data.

- ✓ The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```

# Get summary statistics for numerical columns
print("--- Descriptive Statistics ---")
print(df.describe())

print("--- Key Insights from Statistics ---")
print(f"The average age of a passenger was {df['Age'].mean():.1f} years.")
print(f"The overall survival rate was {df['Survived'].mean():.1%}.")
print(f"Fares ranged from ${df['Fare'].min():.2f} to a whopping ${df['Fare'].max():.2f}.")

```

```

--- Descriptive Statistics ---
      PassengerId  Survived  Pclass     Age  SibSp  \
count    891.000000   891.000000    891.000000  714.000000  891.000000
mean      446.000000     0.383838     2.308642   29.699118    0.523008
std       257.353842     0.486592     0.836071   14.526497    1.102743
min         1.000000     0.000000     1.000000     0.420000    0.000000
25%       223.500000     0.000000     2.000000   20.125000    0.000000
50%       446.000000     0.000000     3.000000   28.000000    0.000000
75%       668.500000     1.000000     3.000000   38.000000    1.000000
max       891.000000     1.000000     3.000000   80.000000    8.000000

```

```

      Parch      Fare
count    891.000000  891.000000
mean       0.381594   32.204208
std       0.806057   49.693429
min        0.000000    0.000000
25%        0.000000    7.910400
50%        0.000000   14.454200
75%        0.000000   31.000000
max         6.000000  512.329200

```

```

--- Key Insights from Statistics ---
The average age of a passenger was 29.7 years.
The overall survival rate was 38.4%.
Fares ranged from $0.00 to a whopping $512.33.

```

Part 3: Visual EDA - Telling Stories with Plots Numbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data into insights. **A Note on**

- ✓ **Plotting Libraries:**
 - * **Matplotlib:** The foundational library, gives you full control over everything.*
 - * **Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

- ✓ Visualization 1: How many survived? A simple `countplot` is the best way to see the distribution of a categorical variable, like our target `Survived`.

```

sns.set_style("whitegrid")

plt.figure(figsize=(8, 6))

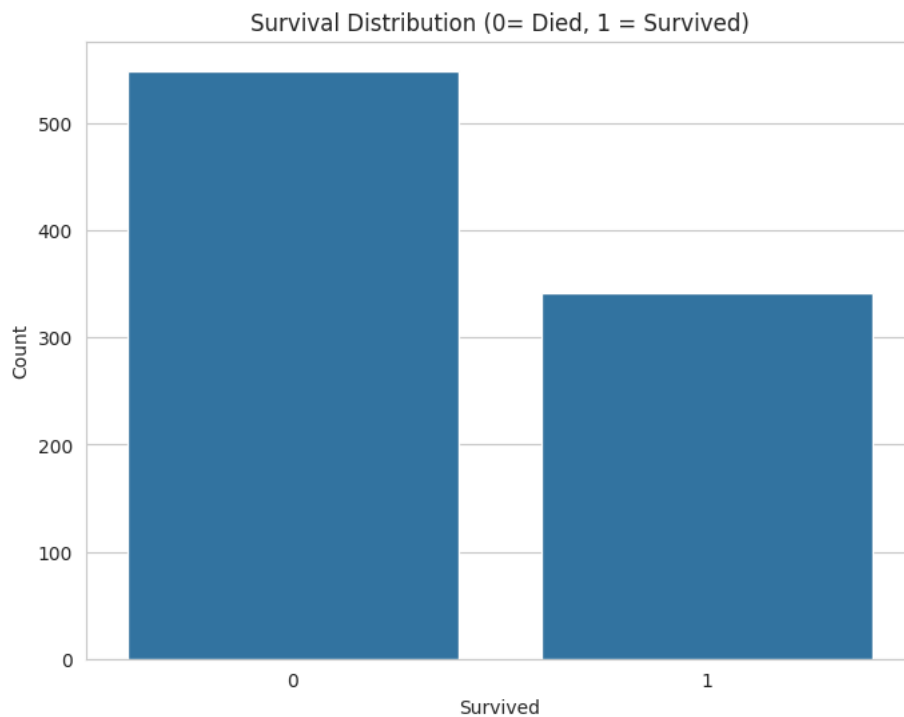
sns.countplot(x="Survived", data=df)

plt.title("Survival Distribution (0= Died, 1 = Survived)")
plt.xlabel("Survived")
plt.ylabel("Count")

plt.show()

```

```
print("Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can sometimes be a chall
```



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Visualization 2: Does passenger class matter for survival? Now we want to see if there's a relationship between two variables. We can use the `hue` parameter in `countplot` to split the bars by another category.

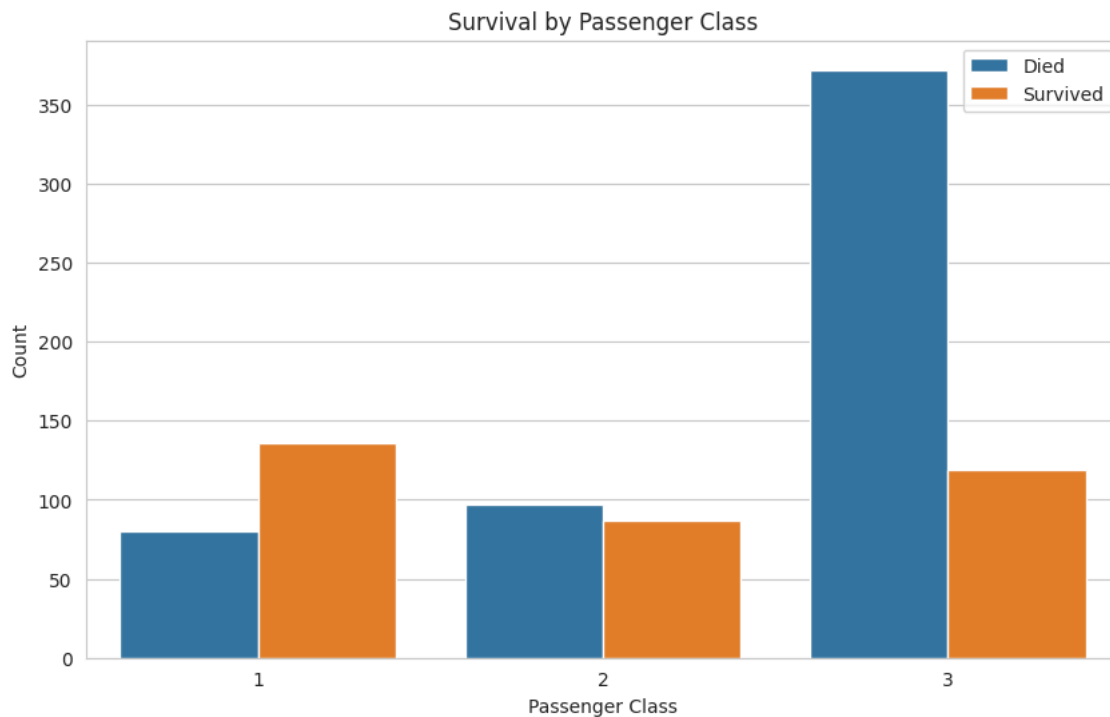
```
plt.figure(figsize=(10,6))
sns.countplot(x="Pclass", hue="Survived", data=df)

plt.title("Survival by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Count")

plt.legend(["Died", "Survived"])

plt.show()

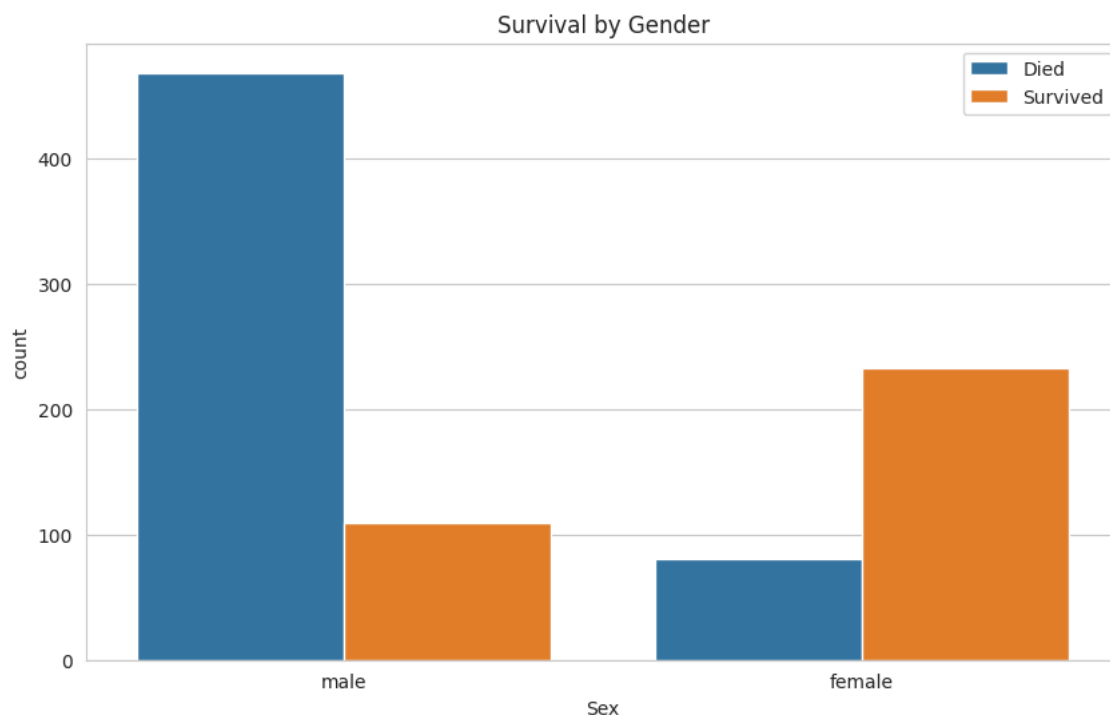
print("Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class
```



Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class passengers.

Visualization 3: What about gender? The 'women and children first' mantra is famous. Let's see if the data supports it.

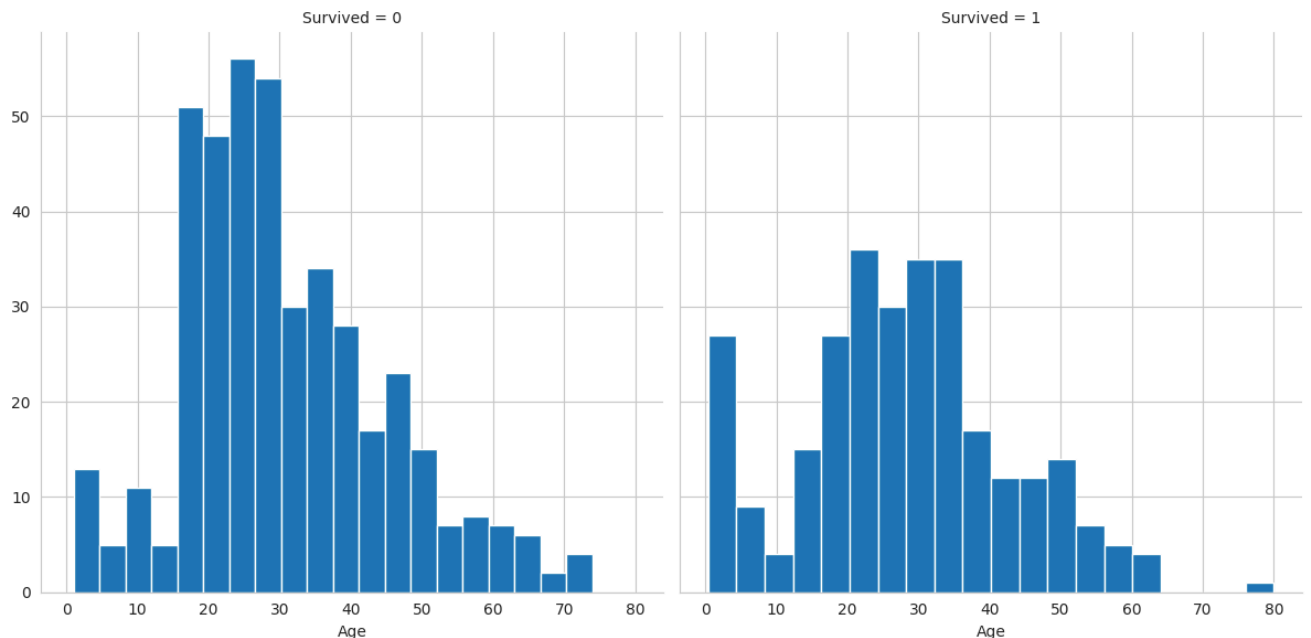
```
plt.figure(figsize=(10, 6))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another very strong pattern.")
```



Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another very strong pattern.

- Visualization 4: How does age play a role? For a continuous variable like `Age`, a histogram is a great way to see its distribution.

```
# A FacetGrid allows us to create multiple plots side-by-side to compare distributions.
# Here, we create one histogram for passengers who died (col='Survived'=0) and
# one for those who survived (col='Survived'=1).
g = sns.FacetGrid(df, col='Survived', height=6)
g.map(plt.hist, 'Age', bins=20)
plt.show()
print("Insight: The age distribution for those who did not survive is centered around the 20-40 age range. For those who survived, the distribution is centered around the 20-40 age range.")
```



Insight: The age distribution for those who did not survive is centered around the 20-40 age range. For those who survived, the distribution is centered around the 20-40 age range.

- Part 4: Student Experimentation **Instructions:** Create your own visualizations to explore other relationships in the data.

Experiment 1: Port of Embarkation1. The `Embarked` column tells you where a passenger boarded the ship (C =

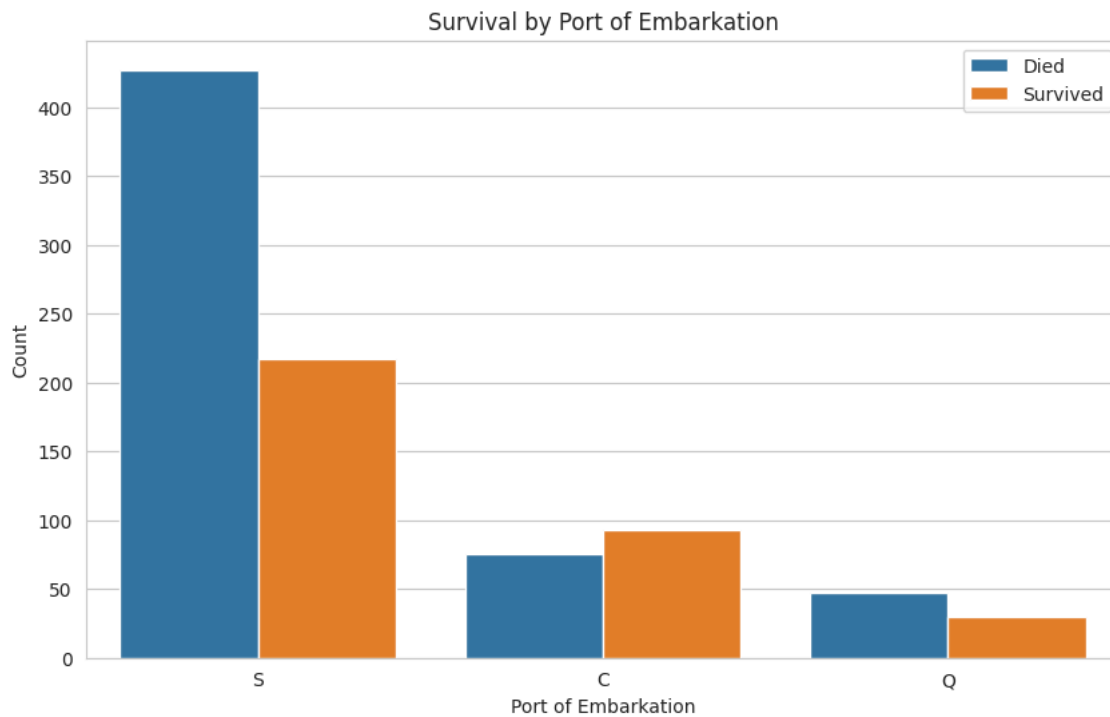
- Cherbourg, Q = Queenstown, S = Southampton).2. Create a `countplot` to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

```
plt.figure(figsize=(10, 6))

sns.countplot(x='Embarked', hue='Survived', data=df)
plt.title( 'Survival by Port of Embarkation')
plt.xlabel( 'Port of Embarkation')
plt.ylabel( 'Count')

plt.legend(['Died', 'Survived'])

plt.show()
```



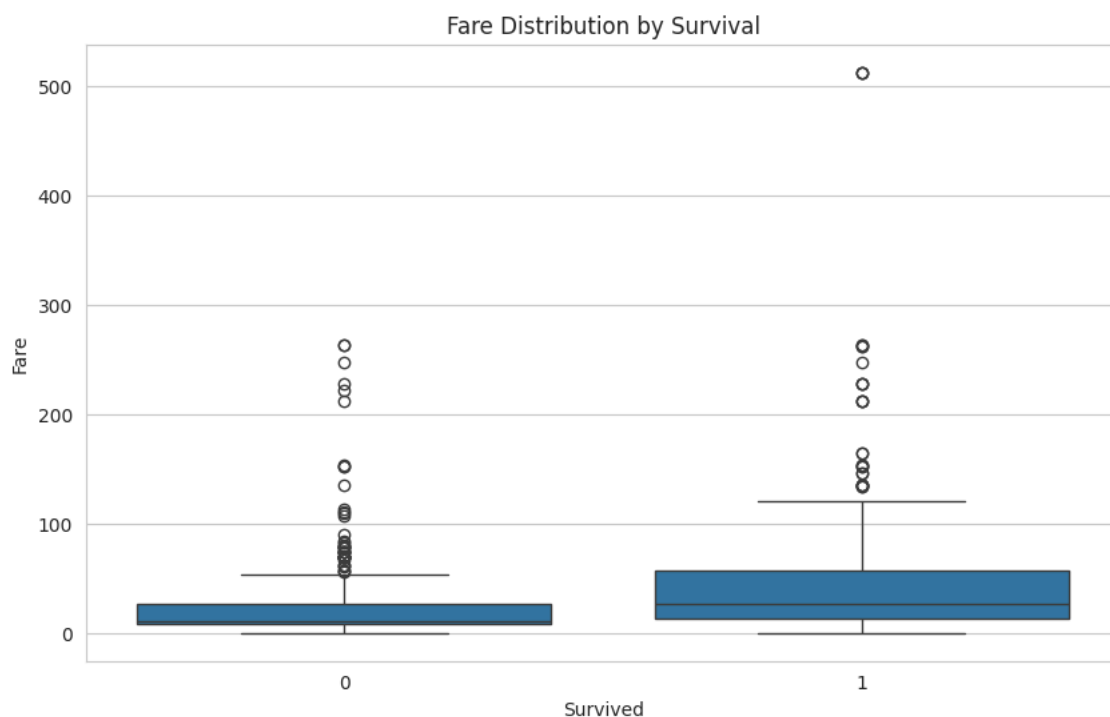
- Experiment 2: Fare vs. Survival1. Fare is a continuous numerical variable.2. A `boxplot` or `violinplot` is a great way to see the distribution of fares for those who survived vs. those who didn't.3. Create one of these plots to compare the Fare distribution by Survived. What does it tell you?

```
plt.figure(figsize=(10, 6))

sns.boxplot(x='Survived', y='Fare', data=df)

plt.title('Fare Distribution by Survival')
plt.xlabel('Survived')
plt.ylabel('Fare')

plt.show()
```





Knowledge CheckInstructions: Answer the following questions in this markdown cell.1.

What is the primary goal of Exploratory Data Analysis (EDA)?2. **Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic? (Describe them in terms of class, gender, and age).**3. **Why is it important to visualize data instead of just looking at summary statistics? What can a plot show you that a number like 'mean' or**